EDUCATION ANALYTICS DASHBOARD

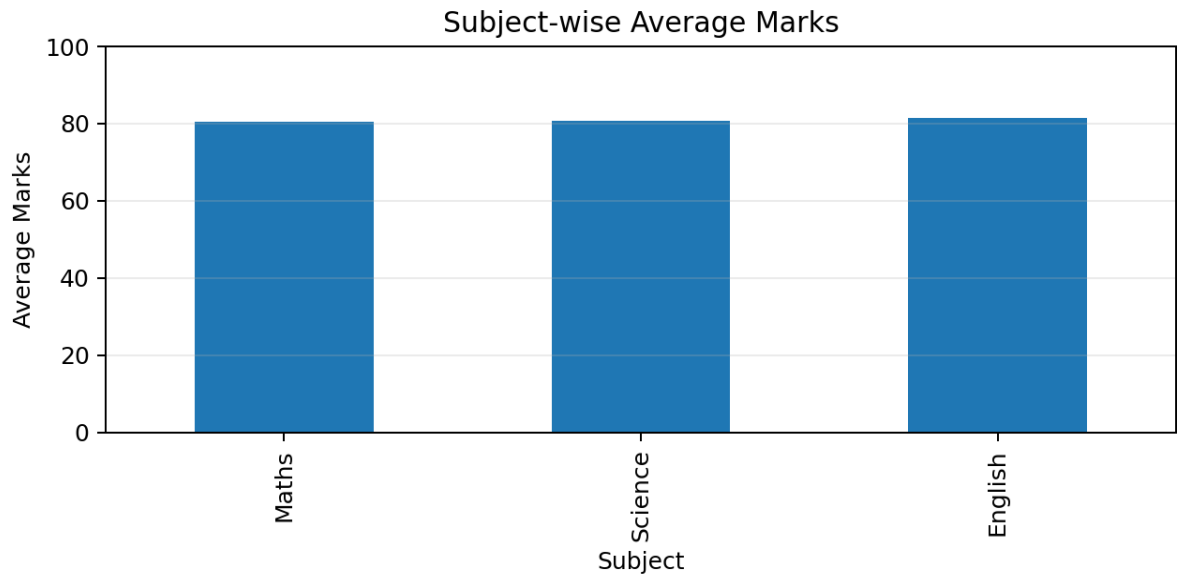
Student marks, attendance, and academic performance analysis

Project objective: summarize student performance using sample classroom data, calculate total and average marks, classify results, and visualize subject averages and attendance.

1. Key Results

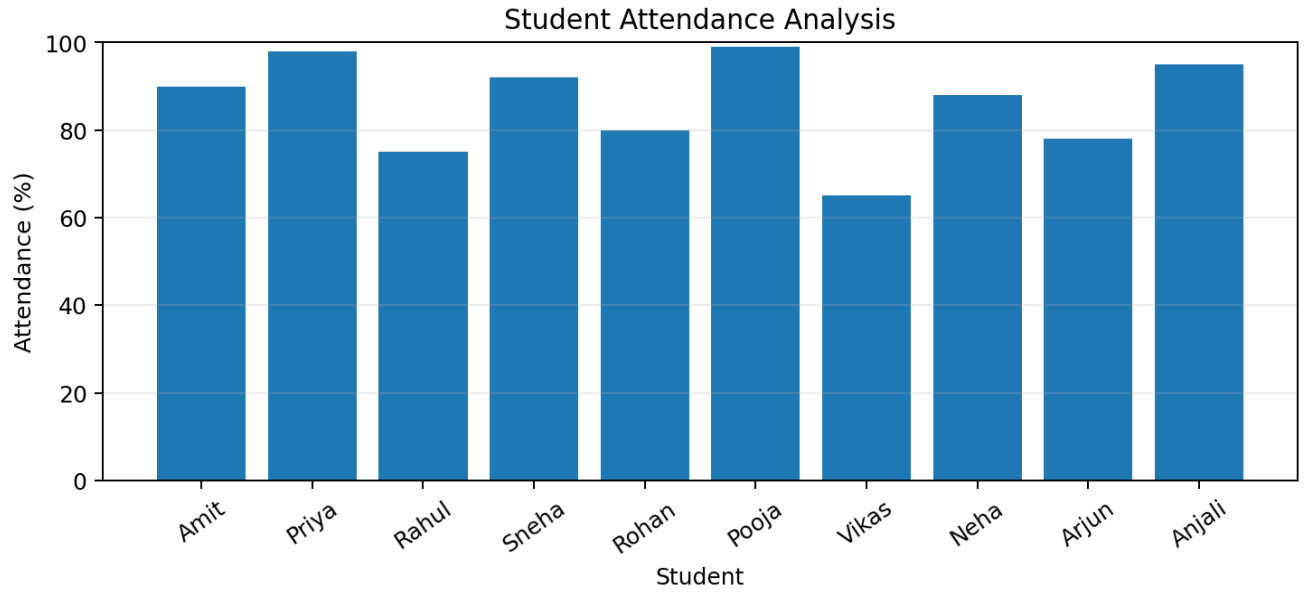
Metric	Output
Total Students	10
Average Class Marks (out of 100)	80.93
Average Attendance	86.00%
Students Passed (rule: average ≥ 35)	10
Students Failed	0

2. Subject-wise Average Marks



Interpretation: compare the average marks across Maths, Science, and English. The chart is generated from the sample dataset used in the notebook.

3. Student Attendance Analysis



Interpretation: attendance varies by student. The chart helps identify students who may need attendance follow-up.

4. Student Performance Table

Student	Maths	Science	English	Attendance	Total	Average	Result
Amit	85	80	78	90%	243	81.00	Pass
Priya	92	95	90	98%	277	92.33	Pass
Rahul	67	70	72	75%	209	69.67	Pass
Sneha	88	85	86	92%	259	86.33	Pass
Rohan	76	78	80	80%	234	78.00	Pass
Pooja	95	92	94	99%	281	93.67	Pass
Vikas	58	60	65	65%	183	61.00	Pass
Neha	81	84	82	88%	247	82.33	Pass
Arjun	73	75	77	78%	225	75.00	Pass
Anjali	90	89	91	95%	270	90.00	Pass

5. Top 3 Students by Average Marks

Student	Average Marks	Attendance	Result
Pooja	93.67	99%	Pass
Priya	92.33	98%	Pass
Anjali	90.00	95%	Pass

6. Python Code Used

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Student": ["Amit", "Priya", "Rahul", "Sneha", "Rohan",
                "Pooja", "Vikas", "Neha", "Arjun", "Anjali"],
    "Maths": [85, 92, 67, 88, 76, 95, 58, 81, 73, 90],
    "Science": [80, 95, 70, 85, 78, 92, 60, 84, 75, 89],
    "English": [78, 90, 72, 86, 80, 94, 65, 82, 77, 91],
    "Attendance": [90, 98, 75, 92, 80, 99, 65, 88, 78, 95]
}
df = pd.DataFrame(data)
df["Total"] = df[["Maths", "Science", "English"]].sum(axis=1)
df["Average"] = df[["Maths", "Science", "English"]].mean(axis=1).round(2)
df["Result"] = df["Average"].apply(lambda x: "Pass" if x >= 35 else "Fail")

print("EDUCATION ANALYTICS DASHBOARD")
print("Total Students:", len(df))
print("Average Class Marks:", round(df["Average"].mean(), 2))
print("Average Attendance:", round(df["Attendance"].mean(), 2), "%")
print("Students Passed:", (df["Result"] == "Pass").sum())
print("Students Failed:", (df["Result"] == "Fail").sum())
display(df)

df[["Maths", "Science", "English"]].mean().plot(
    kind="bar", title="Subject-wise Average Marks")
plt.ylabel("Average Marks")
plt.ylim(0, 100)
plt.show()

df.plot(x="Student", y="Attendance", kind="bar",
        title="Student Attendance Analysis", legend=False)
plt.ylabel("Attendance (%)")
plt.ylim(0, 100)
plt.tight_layout()
plt.show()

print("TOP 3 STUDENTS")
display(df.nlargest(3, "Average")[["Student", "Average", "Attendance", "Result"]])
```